

Test #3 of Erlang

Christian Rinderknecht

7 October 2011

1. Write a function `inter/2` (*set intersection*) such that, if

$$\text{inter}(P, Q) \xrightarrow{+} R,$$

then R is a list made of all the items in common between lists P and Q , without repetition. For example,

$$\text{inter}([a, b, a, c, e], [e, e, b, b, d]) \xrightarrow{+} [b, e].$$

The order of the items in R is left unspecified.

2. Write a function `dist/2` (*distribute*) such that, if

$$\text{dist}(P, Q) \xrightarrow{+} R,$$

then R is the list made of all the possible pairs between the items of P and Q . For example, we can compute a deck of card by distributing the colours over the figures as follows:

$$\begin{aligned} &\text{dist}([\text{heart, diamond, club, spade}], \\ &\quad [2, 3, 4, 5, 6, 7, 8, 9, 10, \text{jack, queen, king, ace}]) \\ &\quad \xrightarrow{+} [\{\text{heart}, 2\}, \{\text{heart}, 3\}, \dots, \{\text{heart}, \text{ace}\}, \\ &\quad \{\text{diamond}, 2\}, \dots, \{\text{spade}, \text{ace}\}]. \end{aligned}$$